We Weren't Scammed!

Reducing Movement Contains the Spread of Pandemic – Agent-based Modeling of Pandemic Spread

Agent-based Modeling of Pandemic Spread confirms the following hypotheses:

- Movement reduction lowers and delays the arrival of the infection peak.
- Across the simulated runs, we do see different pandemic spread trajectory based on the random initialization at t=0, though are consistent overall.
- Depending on the number of parameters and their values, the curve may look similar or different to the SIR curve.

Lessons about Modeling

The key benefit of agent-based modeling is that it allows us to parameterize the mechanisms that population-level models collapse into coarse terms. In SIR model, transmission is governed by beta, which bundles together the movement rate and the avoidance pattern. By modeling each individual, we can represent these components separately. We can then disentangle their contributions to understand which specific behavior or interaction shapes outcomes such as the timing and size of the epidemic peak.



